# Extending DreamCatalyst: Adaptive Guidance, Scene-Aware Localization, and Multi-View Consistency for Text-Driven 3D Editing

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## 1 Thesis at a Glance

**Goal.** Extend DreamCatalyst [1] so that a *single guidance stack* delivers strong text-driven 3D edits and clean background preservation across diverse scenes, auto-adapting to scene type via a scalar *edit-strength* signal read directly from the raw noise predictions, with no per-scene re-tuning.

**Two-phase evaluation.** **Part 1** is a universal 2D guidance/localization/preservation configuration, compared against the DreamCatalyst baseline (Table 1). **Part 2** is an optional 3D voxel-mask cache for multi-view consistency, evaluated *separately* as a module on top of Part 1 (Table 3); this is why the cache does not drive the Part 1 metrics.

**Headline results** (final IEEE configuration, 9 edits on 6 scenes, $n$=3 runs per configuration).

- **Part 1: better preservation on every edit, editability unchanged on average.** One fixed configuration (no per-scene tuning): **SSIM, LPIPS and background LPIPS improve on all 9 edits** beyond the run spread (mean SSIM 0.818 → 0.860, LPIPS 0.289 → 0.233), and background PSNR rises on all 9 (mean +3.8 dB). Whole-image $CLIP_{\text{dir}}$ is unchanged on average (0.158 → 0.160; up on skull, stormtrooper and Batman, down on fangzhou). Inside the edit region, editability rises on 5 edits and falls on 4 (bear, cake, clown, fangzhou), where the baseline changes the region more.
- **Part 2: the voxel cache is metric-neutral.** With nothing else changed, every aggregate mean moves by less than the run-to-run spread; per-edit effects are small and of both signs (strongest positive on stormtrooper). In one Batman run it makes the gloves consistent across two views where the 2D configuration renders bare hands in one of them (Fig. 6).
- **Three transferable findings.** Localization masks must be read *before* guidance amplification; whole-image CLIP-direction must be read with region-restricted metrics, because it rewards background drift toward the target text; and cross-view 3D consistency needs a 3D *memory*, not merely a smoother 2D mask.

**Pipeline.** (i) COLMAP structure-from-motion → (ii) Nerfstudio *nerfacto* reconstruction → (iii) Delta Denoising Score (DDS) [2] editing with an InstructPix2Pix [3] backbone → (iv) optional SDEdit-based refinement. Our contributions live entirely in stage (iii) and operate *after* classifier-free guidance, *before* the DDS gradient is taken. They are modular components that can each be switched off; with all of them off the editor is exactly DreamCatalyst.

**Landscape.** This work connects to recent score-distillation methods on identity preservation and editability, including PDS [6], DaCapo [7], RoMaP [10], Piva [8], SSD [9], and UDS [11]. Our contribution is to combine *mask-based localization* with separate masks for the edit and for preservation, *3D-consistent mask aggregation*, *latent-statistic anchoring*, and *edit-strength-adaptive preservation* inside one modular DreamCatalyst extension.

## 2 Contributions

We organize the contributions into five groups: **G1** guidance amplification (Sec. 2.1), **G2** per-view localization (Sec. 2.2), **G3** cross-view localization (Sec. 2.3), **G4** preservation anchors (Sec. 2.4), **G5** adaptive mechanisms and universal configuration (Sec. 2.5).

### 2.1 G1 — Guidance Amplification

**[A] Adaptive TAG.** TAG [4] decomposes the predicted noise $\boldsymbol{\epsilon}$ into radial $\boldsymbol{\epsilon}_{\parallel}$ (along $\mathbf{v} = \mathbf{z}_t/\|\mathbf{z}_t\|$) and tangential $\boldsymbol{\epsilon}_{\perp}$ components: $\tilde{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_{\parallel} + \eta\,\boldsymbol{\epsilon}_{\perp}, \eta > 1$. Fixed $\eta$ over-amplifies low-noise timesteps. We introduce a concave schedule:

$$\eta(\hat{t}) = 1 + (\eta_{\max} - 1)\,\hat{t}^{\,1/e}, \tag{1}$$

with $1/e \approx 0.368$. The decay naturally aligns with DreamCatalyst's own $w_{\text{DDS}}(\hat{t})$ term.

**[B] Asymmetric TAG.** DDS uses $\Delta\boldsymbol{\epsilon} = \boldsymbol{\epsilon}_{\text{tgt}} - \boldsymbol{\epsilon}_{\text{src}}$. We apply TAG only to the target ($\eta_{\text{tgt}} = \eta(\hat{t})$, $\eta_{\text{src}} = 1$), so the source stays the unmodified reconstruction that DDS subtracts. Because TAG is linear in the prediction, the amplified delta is $\Delta\boldsymbol{\epsilon} + (\eta - 1)\,\boldsymbol{\epsilon}_{\text{tgt},\perp}$: the extra term is the tangential part of the *whole* target prediction, not of the delta.

**[C] STG with a bump schedule.** Following STG [5], a weak UNet pass with self-attention replaced by identity gives $\boldsymbol{\epsilon}_{\text{weak}}$; the amplified prediction is $\boldsymbol{\epsilon}_{\text{stg}} = \boldsymbol{\epsilon}_{\text{full}} +$

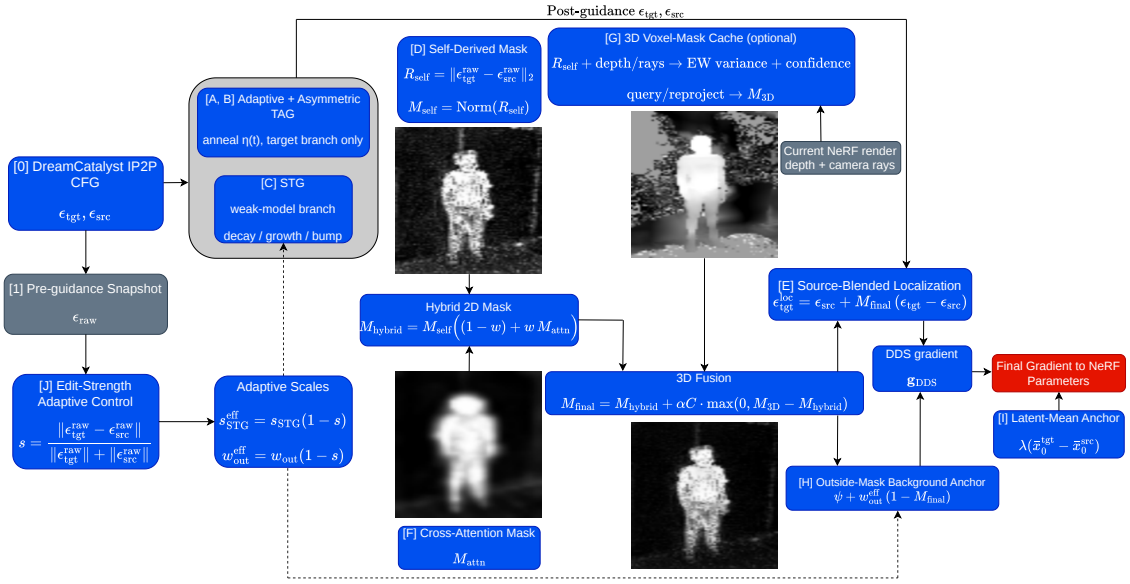



**Figure 1:** Method overview. DreamCatalyst's IP2P CFG is extended with modular guidance, localization, and preservation branches: STG, adaptive/asymmetric TAG, target- and source-instruction masks, cross-attention masking with an agreement floor, 3D voxel-mask aggregation, source-blended localization, outside-mask anchoring, latent-mean anchoring, and adaptive preservation controls. With every component disabled the pipeline is DreamCatalyst.

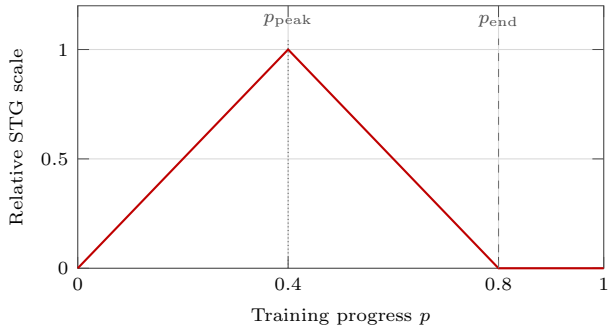



**Figure 2:** STG bump schedule, before the $(1-s)$ attenuation of [J]. STG helps during the mid-training commitment window and is off for the last 20% of training, avoiding per-view freezing on creative edits.

$s\,(\boldsymbol{\epsilon}_{\text{full}} - \boldsymbol{\epsilon}_{\text{weak}})$. TAG and STG act in parallel on the same IP2P prediction, $\tilde{\boldsymbol{\epsilon}}_{\text{tgt}} = \text{TAG}_\eta(\boldsymbol{\epsilon}_{\text{tgt}}) + s^{\text{eff}}_{\text{stg}}(\boldsymbol{\epsilon}_{\text{tgt}} - \boldsymbol{\epsilon}_{\text{weak}})$, with $s_{\text{stg}} = 3.0$ scaled by a **bump** over training progress (zero at the start, peak at 40%, zero again from 80%) and by $(1-s)$ [J]. *Why bump:* late full-strength STG amplifies the *current* NeRF self-attention features, which on creative-transform scenes reflect view-dependent partial-state geometry (e.g. "different armor color per shoulder"). The bump gives mid-phase structural help and leaves the last fifth of training to TAG and localization. Conceptually aligned with DaCapo's temporal edit/preserve separation.

## 2.2 G2 — Localization

**[D] Two instruction masks: one for the edit, one for preservation.** With $f$ the image-guided part of the IP2P CFG, the raw DDS delta splits exactly into what the instruction still *demands* of the current render and the *displacement* the render has already made:

$$\Delta\boldsymbol{\epsilon}^{\text{raw}} = \underbrace{s_{\text{txt}}\big(\boldsymbol{\epsilon}_{\text{txt}} - \boldsymbol{\epsilon}_{\text{img}}\big)(z_t)}_{\text{demand}} + \underbrace{f(z_t) - f(z^{\text{src}}_t)}_{\text{displacement}}.$$

The thesis self-mask used $\|\Delta\boldsymbol{\epsilon}^{\text{raw}}\|$, so it rose wherever the render had drifted, wanted or not, and then admitted more edit force there. The final design separates the roles.

- **Edit mask $M_{\text{tgt}}$**: the demand alone, i.e. the per-pixel norm of $\boldsymbol{\epsilon}_{\text{txt}} - \boldsymbol{\epsilon}_{\text{img}}$ of the target pass (the instruction contrast of LatentEditor [16]), $p_5$–$p_{95}$ normalized, smoothed by a per-view EMA with a memory of 5 visits of that view ($\beta = e^{-1/5}$), sharpened ($\gamma = 1.2$) and blurred. It follows the edit as it forms, so new structure can receive edit force. (The thesis EMA, $\beta = 1 - 1/(2N_{\text{cam}})$, advanced only on visits of its own view, so on the 315-view person scene each view's mask kept ≈99% of its first visit until the end of the run.)
- **Preservation mask $M_{\text{src}}$**: the same contrast on the *source* latent noised at $t = 500$, averaged over 4 noise draws (DiffEdit's recipe [12]), computed once per view and cached. The edit cannot change it, so a drifting region can never relax its own restoring force [H].

Both masks, like the edit strength [J], are read *before* TAG or STG modify the predictions, so they cannot inherit the amplification they are meant to restrain, consistent with DiffEdit's pre-guidance mask.

**[E] Source-blended localization.** Inspired by **LatentEditor** [16], we blend inside the DDS score rather than



Rendered Self-mask CA mask $M_{\text{attn}}$ Fused mask

**Figure 3:** Mask hierarchy, from a run of the thesis design (DDS-delta self-mask, product fusion). The self-derived mask gives fine-grained localization and the cross-attention mask a semantic prior. The final design builds the self-mask from the instruction contrast [D] and bounds the attention veto with the source-instruction floor [F].

post-hoc:

$$\boldsymbol{\epsilon}^{\text{loc}}_{\text{tgt}} = \boldsymbol{\epsilon}_{\text{src}} + M \odot (\tilde{\boldsymbol{\epsilon}}_{\text{tgt}} - \boldsymbol{\epsilon}_{\text{src}}), \tag{2}$$

so outside-mask regions fall back to the source branch before the DDS difference is computed. The blend acts on the guided prediction $\tilde{\boldsymbol{\epsilon}}_{\text{tgt}}$, so TAG and STG are localized together with the edit, and it masks only the DDS term: DreamCatalyst's preservation term stays unmasked and keeps pulling toward the source outside the mask. FoI [17] and ZONE [18] support the broader idea of moving localization *inside* the editing process.

**[F] Cross-attention prior with an agreement floor.** Following **Prompt-to-Prompt** [14], **DAAM** [15], and **LEDITS++** [13], we aggregate target-token cross-attention from UNet up-blocks $[1, 2]$ into $M_{\text{attn}}$. Target tokens are auto-derived from the prompts (target-only words minus stopwords and source words), with no manual per-scene keyword overrides. The product $M_{\text{tgt}} M_{\text{attn}}$ can only subtract support. At full strength from the first step it keeps the edit off the background, but it also vetoes parts of the requested edit that the token does not attend to: with the product alone, a remnant of the source hand remained beside the stormtrooper's glove. The thesis ramped the CA weight in late, which avoids that veto early but leaves the early, high-noise steps unlocalized. The final design keeps CA at full strength and bounds what it may veto:

$$M = \max\big(M_{\text{tgt}}\, M_{\text{attn}},\ \min(M_{\text{tgt}}, M_{\text{src}})\big). \tag{3}$$

Attention keeps the final say only where the two instruction masks disagree (structure beyond the source silhouette, spill the source does not support). $M$ never exceeds $M_{\text{tgt}}$, the floor needs two measurements on different latents to agree, and it adds no hyperparameter.

## 2.3 G3 — 3D Cross-View Localization

**[G] 3D voxel-mask cache for cross-view localization.** The masks above are computed from one rendered view at a time. This is the core limitation behind the *clown* failure case: view 1 may strongly edit an arm while view 165 does not, and no per-view EMA can make those two rays talk to each other. To add a lightweight 3D consistency prior, we lift the current 2D edit mask into a non-parametric world-space voxel cache.

For a pixel $(u, v)$ with rendered depth $D(u, v)$ and Nerfstudio world ray $(\mathbf{o}, \mathbf{d})$, we backproject

$$\mathbf{x}(u, v) = \mathbf{o} + D(u, v)\, \mathbf{d}. \tag{4}$$

The point is discretized into a voxel index $\mathbf{q}(\mathbf{x})$. If the ray terminates on a surface (rendered accumulation above 0.5), the voxel stores a running average of the *raw* target-instruction magnitude $M_{\text{raw}}$ of [D], divided by its per-frame 95th percentile (the rank-normalized edit mask would destroy the absolute foreground/background contrast, and it already contains the attention restriction and the cache's own additions):

$$V_{\mathbf{q}} \leftarrow \beta_{\text{EMA}} V_{\mathbf{q}} + (1 - \beta_{\text{EMA}}) M_{\text{raw}}(u, v), \tag{5}$$

with $\beta_{\text{EMA}} = 1 - 1/(0.5\, N_{\text{cam}})$. A voxel's first observation is copied, so a long memory would leave the read-back "consensus" mostly the first observer's value throughout the fusion ramp. With the factor 0.5, a voxel seen by 40% of 315 cameras has about 9 effective observations behind its value when fusion starts and about 160 when it reaches full strength (about 2 and 4 with the thesis factor 2). At later views the same cache is queried by backprojecting their pixels into the same world grid, and the queried value is fused with the 2D mask $M$ of [F] through a *positive-only* correction:

$$M_{\text{final}} = M + \alpha\, C\, G\, \max\big(M_{\text{3D}} - M,\, 0\big), \tag{6}$$

where $\alpha$ is a warmup-controlled blend, $C$ is the per-voxel confidence (observation count, exponentially weighted cross-view variance, angular diversity and cached mass) and $G = \max(M_{\text{attn}}, M_{\text{tgt}})\, M_{\text{attn}}$ is a semantic gate. The factor $M_{\text{attn}}$ matters: the cache is built from the same contrast as $M_{\text{tgt}}$, so $M_{\text{tgt}}$ alone must not be able to authorize an addition. The cache adds edit support where the 3D consensus exceeds the 2D evidence, and never subtracts. A subtractive (bidirectional) variant, which also pulled the 2D mask down toward the 3D consensus, was evaluated and discarded: the averaged-across-views cache value systematically sits below the sharp 2D peaks, so the subtraction eroded genuine fine detail without improving consistency. The design is inspired by Semantic-NeRF-style rendering of scene-level fields [19] and by panoptic/semantic lifting of noisy 2D predictions into multi-view-consistent 3D representations [20]. InterGSEdit pursues a related cross-view-consistent editing goal in 3DGS via inter-view attention [21]; our cache is a lightweight, non-parametric NeRF-friendly alternative. In this thesis, the novelty is the *online supervision signal*: the 3D mask is built from the diffusion model's own instruction contrast during editing, not from precomputed segmentation labels.

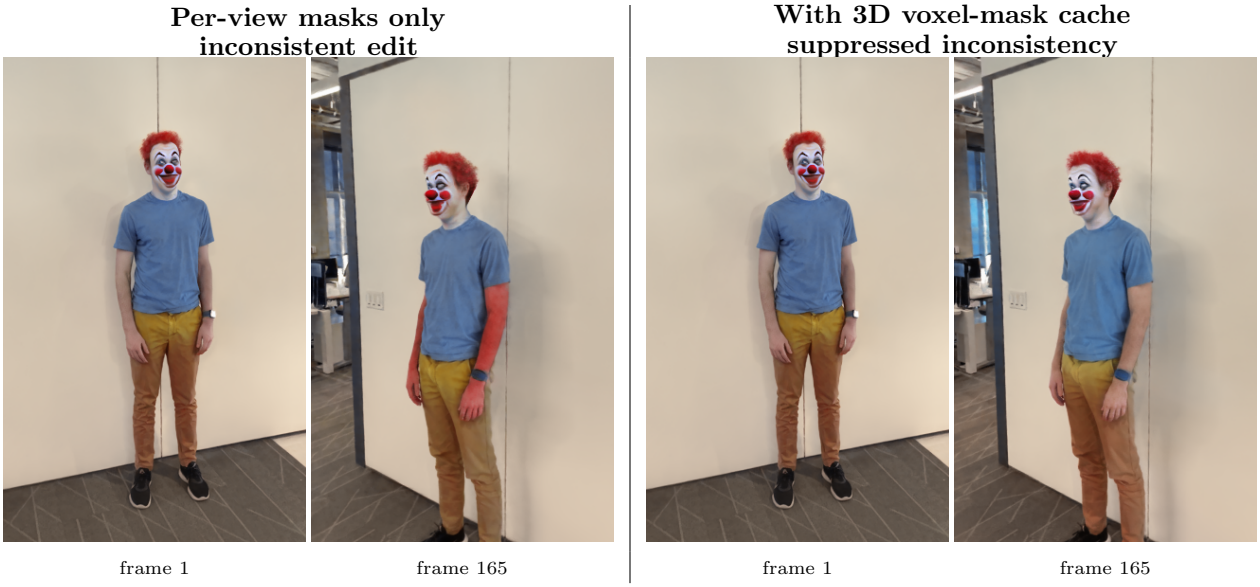



**Figure 4: Qualitative motivation for the 3D voxel-mask cache** (thesis runs, earlier configuration). Without a 3D mask prior, the clown edit became view-dependent: one training view accumulated arm/costume edits that another view did not. With the cache, the same two views agreed. A cross-view example from the final runs is Fig. 6.

### 2.4 G4 — Preservation Anchors

**[H] Outside-mask background anchor.** The DreamCatalyst preservation term $e^{\hat{\psi}}\psi\,(x_0^{\text{tgt}} - x_0^{\text{src}})$ is strengthened where the instruction does not apply: preserve$(x) = \psi + w_{\text{out}}\,(1-s)\,(1 - M_{\text{src}}(x))$, with $w_{\text{out}} = 0.25$. It reads the fixed source mask, not the edit mask, so the brake is a fixed potential: a region that drifts toward the target cannot relax its own restoring force, and in Part 2 a cache false positive cannot gain edit force and lose preservation at the same time. Conceptually aligned with RoMaP's [10] spatial regularization.

**[I] Latent-mean anchor.** With TAG active we observed a systematic brightness and saturation drift of the per-channel means in VAE-latent space. We add a simple analytic bias to the final gradient:

$$\mathbf{g} \leftarrow \mathbf{g} + \lambda\,(\overline{x}_0^{\text{tgt}} - \overline{x}_0^{\text{src}}), \tag{7}$$

where $\overline{x}_0 = \text{mean}(x_0, \text{dim}{=}(H, W))$ is the per-channel spatial mean. This regularizes channel-mean drift directly in latent statistics. Spirit follows Piva [8] (explicit identity term) and SSD [9] (source-anchored stabilization). Empirical optimum on face: $\lambda = 0.005$.

### 2.5 G5 — Adaptive Mechanisms & Universal Configuration

**Core idea.** The same hyperparameter should not be optimal for identity-preserving scenes (face) and creative-transform scenes (stormtrooper). Instead of re-tuning per scene, we make preservation *self-adjust* based on a cheap proxy of scene type: the *edit strength* $s \in [0, 1]$ read directly from the raw noise predictions.

**[J] Edit-strength-adaptive preservation.** From the clean pre-guidance DDS snapshots $\boldsymbol{\epsilon}_{\text{tgt}}$ and $\boldsymbol{\epsilon}_{\text{src}}$ (captured before STG and TAG, like the masks of [D]), we define

$$s = \frac{\|\boldsymbol{\epsilon}_{\text{tgt}}^{\text{raw}} - \boldsymbol{\epsilon}_{\text{src}}^{\text{raw}}\|}{\|\boldsymbol{\epsilon}_{\text{tgt}}^{\text{raw}}\| + \|\boldsymbol{\epsilon}_{\text{src}}^{\text{raw}}\|} \in [0, 1],$$

which is bounded by the triangle inequality and invariant to a common rescaling of the two predictions; it grows with the relative disagreement of the two branches. We scale both the background anchor and the STG scale by $(1 - s)$:

$$w_{\text{out}}^{\text{eff}} = w_{\text{out}}\,(1 - s), \tag{8}$$

$$s_{\text{stg}}^{\text{eff}} = s_{\text{stg}}\,(1 - s). \tag{9}$$

A single $(w_{\text{out}}, s_{\text{stg}})$ pair serves every scene; bold edits loosen both automatically, subtle edits keep both near full strength.

**Universal configuration.** Combining the evaluated components (A–F and H–J) yields a single configuration tested across all 9 edits without per-scene tuning (see Sec. 3.1); the 3D voxel cache (G) is evaluated separately as an optional module (Sec. 3.2).

## 3 Experiments

**Setup.** NeRF (*nerfacto*) + DreamCatalyst-DDS with IP2P as the editor, 3000 edit iterations, fixed $s_{\text{txt}}$=7.5 and $s_{\text{img}}$=1.5 for fair comparison with the reference paper. Three configurations run the same code: the DreamCatalyst baseline (every module off), Part 1 (2D) and Part 2 (3D, Part 1 plus the voxel cache, nothing else changed). Metrics: CLIP-direction ↑ (editability), CLIP-image ↑, SSIM ↑ and LPIPS ↓ (preservation), multi-view cosine

similarity ↑, the custom *EditMaskVariance_3D* ↓ for Part 2, and PIE-Bench-style region-restricted metrics computed per view against a CLIPSeg region segmented from the *source* frames only: editability inside the region (CLIP-dir-local), background LPIPS/PSNR outside it, and the edit localization ratio (ELR). All numbers are means over $n$=3 runs on a fixed reconstruction checkpoint per scene. A difference counts only when it exceeds the combined run spread $\sqrt{\sigma_A^2 + \sigma_B^2}$ (**green** in the tables); the run-to-run $\sigma$ of CLIP-direction, averaged over edits, is 0.005 (DC), 0.006 (2D) and 0.003 (3D).

**Edits.** Nine edits on six scenes: face (Tolkien elf, skull), fangzhou (knight helmet), yuseung (Batman, Joker), person (clown, stormtrooper), bear (panda), and cake (red velvet to chocolate, a CO3D sequence). Compared with the thesis set, Einstein is no longer evaluated, and skull and cake are new.

## 3.1 Part 1: Universal Configuration vs. DreamCatalyst

- **Preservation improves on every edit.** SSIM, LPIPS and background LPIPS improve on all 9 edits beyond the run spread, background PSNR on all 9 (8 beyond), and CLIP-I on 8 (clown is the exception). The gains are largest where the baseline changes the background most: cake (background PSNR 17.7→28.7 dB) and fangzhou (19.1 → 25.7 dB). On average the change outside the edit region falls by 44%.
- **Whole-image editability is unchanged on average** (0.158→0.160): up beyond the spread on skull (+19%), stormtrooper (+13%) and Batman (+8%), down on fangzhou (0.208→0.189), a tie on the other five.
- **Inside the edit region the picture splits.** CLIP-dir-local rises beyond the spread on elf, skull, stormtrooper, Batman and Joker, falls beyond it on bear (marginally), cake and clown, and is lower within the spread on fangzhou. Our configuration changes the region less on every edit (mean in-region change 0.195 vs. 0.271); on the second group the baseline's stronger change also scores higher.
- **The region metrics expose background drift.** On cake the baseline's background moved toward the target text (CLIP-dir-bg 0.054 vs. 0.007), which whole-image CLIP-direction rewards.
- **The renders show the same pattern** (Fig. 5). The baseline darkens the grass around the cake, turns the fangzhou sweater metallic and smooths away the tiled wall, and turns the elf's hair green; ours changes the cake, adds the helmet and the pointed ear, and keeps the rest.

**Final configuration (fully general; no keywords or segmentation masks):** adaptive asymmetric TAG ($\eta_{\text{tag}} = 1.2$); STG (scale 3.0, bump peaking at 40% and off from 80%, edit-strength-adaptive); edit mask $M_{\text{tgt}}$ from the target-instruction contrast (per-view EMA, memory 5 visits, $\gamma = 1.2$); preservation mask $M_{\text{src}}$ from the source-instruction contrast ($t = 500$, 4 noise draws, cached per view); cross-attention prior at full strength from the first step ($\gamma = 1.2$, up-blocks $[1, 2]$, auto-derived tokens) with the agreement floor; source-blended localization; an outside-mask anchor (0.25) on $1 - M_{\text{src}}$ with edit-strength-adaptive scaling; and a latent-mean anchor (0.005). Part 2 enables the voxel cache and changes nothing else.

## 3.2 Part 2: Multi-View Consistency via the 3D Voxel Cache

**Part 2.** The universal configuration edits each view independently, so a region can be edited in one camera and preserved in another (in the thesis runs, the *clown*'s arm, Fig. 4). The 3D voxel cache (contribution G) is an *optional* module that lifts the per-view edit evidence into a shared world grid and reprojects it to later views, so the edit *decision* becomes shared across cameras.

Evaluated on top of Part 1 with nothing else changed ($n$=3, nine edits; Table 3), the cache is **metric-neutral in aggregate**: every aggregate mean, including the region-restricted metrics and *EditMaskVariance_3D*, changes by less than the typical run-to-run spread. Per edit, 19 of the 99 comparisons (eleven metrics, nine edits) separate beyond the spread, 11 in favour of the cache and 8 against: six of the favourable ones are on stormtrooper (CLIP-direction, CLIP-I, LPIPS, MV Cos and both background metrics) and three of the unfavourable ones on clown (localization ratio and both background metrics). MV Cos is higher with the cache on 7 of 9 edits (3 beyond the spread) and lower beyond the spread on skull. **In the renders** (Fig. 6), one Batman run shows the kind of cross-view difference the cache targets: without it the hands are gloved in view 1 and bare in view 71; with it they are gloved in both, as in DreamCatalyst. The same figure shows that the stormtrooper's ghost hand persists faintly with and without the cache.

**Why the readback matters.** A nearest-voxel lookup inherits the grid resolution, so a mask that is smooth in 3D returns to the 2D score with voxel-sized steps, and with speckle wherever adjacent voxels disagree or were never written. Plain trilinear interpolation removes the steps but averages in voxels the cache has never observed, pulling the mask toward zero exactly where view coverage is thinnest. The implementation therefore masks each of the eight corner weights by an `observed` flag and renormalizes by the surviving weight,

$$M_{\text{3D}}(x) \;=\; \frac{\sum_{k=1}^{8} w_k \, o_k \, V_k}{\sum_{k=1}^{8} w_k \, o_k},$$

falling back to a neutral value only when no surrounding voxel has been seen at all. The cross-view variance that gates trust is read over the same eight corners (corners seen by at least two views, within-voxel variance only), so the agreement gate is as continuous as the value it gates.

**Table 1: Part 1, whole-image metrics: DreamCatalyst (DC) → universal configuration**, mean over $n$=3 runs, one fixed configuration (no per-scene tuning). Better value in **green** when the gap exceeds the combined run spread; the last rows count the edits on which each side is better (beyond the spread in parentheses).

| Edit | $\text{CLIP}_{\text{dir}}$ ↑ | CLIP-I ↑ | SSIM ↑ | LPIPS ↓ | MV Cos ↑ |
|---|---|---|---|---|---|
| Face / Tolkien elf | 0.128 → 0.134 | 0.598 → 0.600 | 0.818 → **0.837** | 0.254 → **0.213** | 0.918 → 0.924 |
| Face / skull | 0.086 → **0.103** | 0.587 → **0.598** | 0.796 → **0.824** | 0.272 → **0.240** | 0.918 → **0.923** |
| Fangzhou / knight helmet | **0.208** → 0.189 | 0.713 → **0.774** | 0.772 → **0.841** | 0.331 → **0.266** | **0.956** → 0.939 |
| Yuseung / Batman | 0.153 → **0.165** | 0.633 → **0.658** | 0.868 → **0.889** | 0.268 → **0.245** | 0.927 → **0.934** |
| Yuseung / Joker | 0.172 → 0.176 | 0.645 → **0.662** | 0.912 → **0.923** | 0.239 → **0.221** | 0.934 → 0.936 |
| Person / clown | 0.138 → 0.137 | **0.775** → 0.764 | 0.926 → **0.932** | 0.223 → **0.200** | 0.913 → 0.910 |
| Person / stormtrooper | 0.163 → **0.185** | 0.637 → **0.684** | 0.796 → **0.838** | 0.336 → **0.297** | **0.916** → 0.904 |
| Bear / panda | 0.177 → 0.175 | 0.737 → **0.779** | 0.694 → **0.764** | 0.282 → **0.235** | 0.900 → 0.899 |
| Cake / chocolate | 0.201 → 0.175 | 0.767 → **0.872** | 0.776 → **0.895** | 0.393 → **0.177** | 0.929 → **0.958** |
| **Mean** | 0.158 → 0.160 | 0.677 → 0.710 | 0.818 → 0.860 | 0.289 → 0.233 | 0.923 → 0.925 |
| Ours better | 5 (3) | 8 (7) | 9 (9) | 9 (9) | 5 (3) |
| DC better | 4 (1) | 1 (1) | 0 (0) | 0 (0) | 4 (2) |

**Table 2: Part 1, region-restricted metrics**, same layout as Table 1. For CLIP-dir-bg the better value is the one closer to zero (the background did not move toward the target text).

| Edit | CLIP-dir-local ↑ | CLIP-dir-bg → 0 | ELR ↑ | Bg LPIPS ↓ | Bg PSNR ↑ |
|---|---|---|---|---|---|
| Face / Tolkien elf | 0.134 → **0.149** | −0.007 → −0.013 | 0.654 → **0.686** | 0.162 → **0.140** | 25.37 → **27.89** |
| Face / skull | 0.099 → **0.121** | −0.014 → −0.012 | 0.705 → **0.745** | 0.172 → **0.148** | 25.46 → **27.72** |
| Fangzhou / knight helmet | 0.222 → 0.205 | 0.081 → 0.088 | 0.469 → **0.557** | 0.200 → **0.151** | 19.14 → **25.73** |
| Yuseung / Batman | 0.153 → **0.160** | −0.002 → −0.005 | **0.791** → 0.763 | 0.152 → **0.147** | 30.02 → **30.80** |
| Yuseung / Joker | 0.173 → **0.182** | −0.011 → −0.009 | **0.730** → 0.694 | 0.150 → **0.145** | 30.83 → 30.92 |
| Person / clown | **0.157** → 0.142 | −0.039 → −0.035 | 0.664 → **0.754** | 0.130 → **0.112** | 27.41 → **33.16** |
| Person / stormtrooper | 0.177 → **0.185** | −0.063 → **-0.046** | 0.801 → 0.806 | 0.139 → **0.120** | 28.41 → **31.31** |
| Bear / panda | **0.203** → 0.200 | 0.010 → 0.013 | 0.573 → **0.599** | 0.120 → **0.106** | 21.22 → **23.52** |
| Cake / chocolate | **0.249** → 0.213 | 0.054 → **0.007** | 0.550 → **0.639** | 0.293 → **0.110** | 17.74 → **28.68** |
| **Mean** | 0.174 → 0.173 | 0.001 → −0.001 | 0.660 → 0.694 | 0.169 → 0.131 | 25.07 → 28.86 |
| Ours better | 5 (5) | 5 (2) | 7 (6) | 9 (9) | 9 (8) |
| DC better | 4 (3) | 4 (0) | 2 (2) | 0 (0) | 0 (0) |

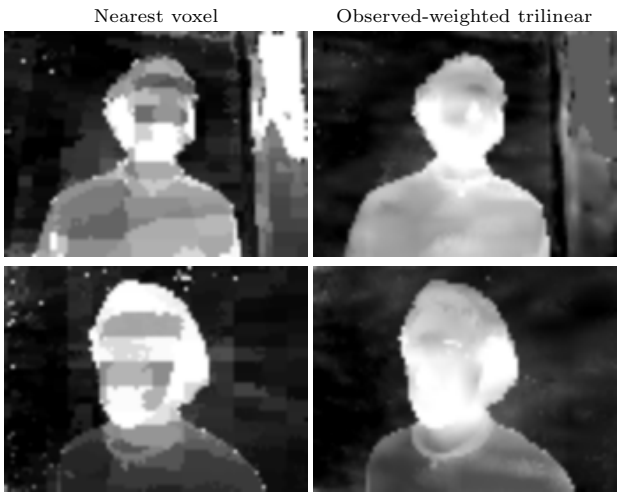



**Figure 7: Voxel readback decides whether a 3D-consistent mask survives as a usable 2D mask** (thesis development run, *elf*, cache contents fixed, only the readback differs). Nearest-voxel lookup (left) returns the grid quantized: block edges across the subject and salt-and-pepper speckle on the background. Observed-weighted trilinear (right) returns a continuous mask with a smooth boundary.

The cache is non-parametric and builds its supervision online from the diffusion model's own instruction contrast, never from precomputed segmentation. It inherits the quality of the per-view mask, and on broad-recolouring edits it needs the variance-based agreement gate so that it does not also make a *coherent-but-unwanted* edit more consistent. The mechanisms that did *not* work in the thesis, a peak-hold variance latch and active suppression of contested regions, are documented as negative results.

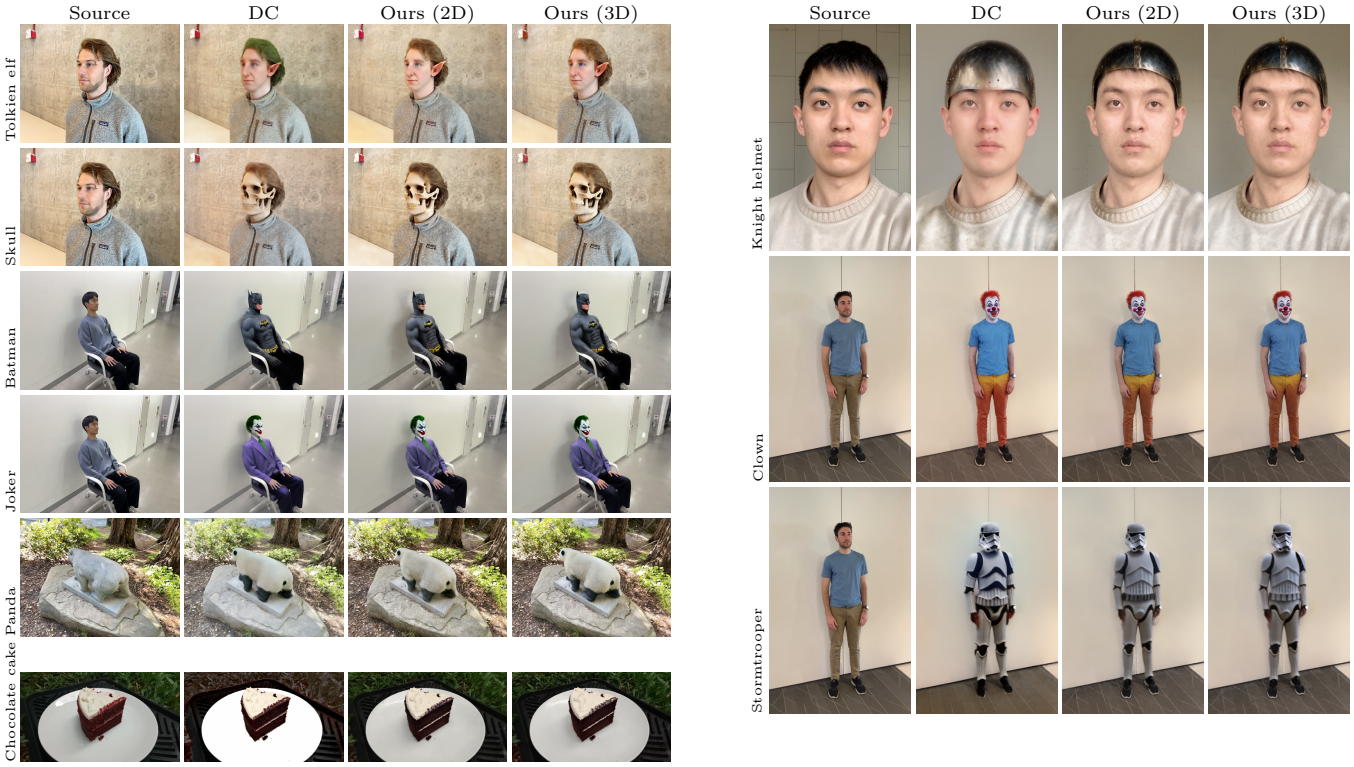



**Figure 5: Qualitative results across all nine edits.** Source image, the DreamCatalyst baseline, our universal configuration (Part 1) and the same configuration with the 3D voxel cache (Part 2), from the same camera within each row, one run of each configuration.

## 4 Methodological Findings

**(1) Masks must be read before guidance amplification.** TAG's tangential amplification and STG's structural extrapolation enlarge the target prediction where they are most active; a mask built after them would grow there and feed the amplification back into its own gate. All masks and the edit strength are therefore read from the IP2P passes before TAG and STG act, as DiffEdit [12] builds its mask before any editing guidance.

**(2) Whole-image editability must be read with region-restricted metrics.** Whole-image CLIP-direction asks whether the picture moved toward the text, not where. On cake the baseline's background moved toward "chocolate" (CLIP-dir-bg 0.054 vs. 0.007), and on fangzhou its whole-image score is higher while its background PSNR is 6.6 dB lower. The region-restricted pair (CLIP-dir-local inside the region, background LPIPS/PSNR outside it) separates the edit from the leakage.

**(3) 3D consistency requires a 3D memory, not only a smoother 2D mask.** A per-view EMA can reduce temporal flicker, but it cannot make two different cameras agree. The voxel cache stores edit evidence at an approximate 3D point and reuses it when another view observes that point. It does not solve semantic ambiguity by itself; it turns a per-view localization signal into a more view-consistent one.

## 5 Limitations & Planned Extensions

**The voxel cache does not improve the aggregate metrics.** In the final runs every aggregate mean stays within the run-to-run spread, and the per-edit effects are small and of both signs. It inherits the quality of the per-view mask: if the 2D mask is semantically wrong, the cache can make that wrong signal more consistent.

**Preservation is not free on every edit.** On bear, cake, clown and fangzhou the baseline scores higher in region-restricted editability while changing the edit region more, and on fangzhou its whole-image editability is also higher. The metrics do not settle which outcome a user would prefer there.

**The STG schedule shape is fixed, not inferred.** Edit-strength-adaptive scaling adjusts the STG *magnitude*, but the bump shape is used for every scene, although the thesis runs suggested that the best shape depends on the edit's trajectory. Making the shape itself driven by $s$ is a small extension.

**Shrinking-silhouette edits.** In the thesis runs the stormtrooper's gloved hand, smaller than the person's original hand, left a translucent ghost-hand residue. The agreement floor [F] and the source-mask anchor [H] target it, but in the final runs a faint remnant of the source fingers still sits beside the glove in one of the two views shown,

**Table 3: Part 2: universal configuration (2D, cache off) → with the 3D voxel cache**, mean over $n$=3 runs, nothing else changed. $EMV_{3D}$ is the custom mask-level 3D-consistency metric (lower is nominally better, though confounded by edit strength). Better value in **green** only when the gap exceeds the combined run spread.

| Edit | $\text{CLIP}_{\text{dir}}$ ↑ | CLIP-I ↑ | SSIM ↑ | LPIPS ↓ | MV Cos ↑ | $\text{EMV}_{\text{3D}}$ ↓ |
|---|---|---|---|---|---|---|
| Face / Tolkien elf | 0.134 → 0.136 | 0.600 → 0.599 | 0.837 → 0.838 | 0.213 → 0.212 | 0.924 → **0.929** | **0.0209** → 0.0223 |
| Face / skull | 0.103 → 0.100 | 0.598 → 0.600 | 0.824 → 0.821 | 0.240 → 0.238 | **0.923** → 0.917 | 0.0182 → 0.0169 |
| Fangzhou / knight helmet | 0.189 → 0.188 | 0.774 → 0.769 | **0.841** → 0.838 | 0.266 → 0.269 | 0.939 → **0.942** | 0.0175 → 0.0175 |
| Yuseung / Batman | 0.165 → 0.160 | 0.658 → 0.653 | 0.889 → 0.889 | 0.245 → 0.246 | 0.934 → 0.936 | 0.0104 → 0.0105 |
| Yuseung / Joker | 0.176 → 0.175 | 0.662 → 0.657 | 0.923 → 0.920 | 0.221 → 0.226 | 0.936 → 0.937 | 0.0208 → 0.0201 |
| Person / clown | 0.137 → 0.137 | 0.764 → **0.776** | 0.932 → 0.934 | 0.200 → 0.200 | 0.910 → 0.912 | 0.0059 → 0.0063 |
| Person / stormtrooper | 0.185 → **0.189** | 0.684 → **0.698** | 0.838 → 0.840 | 0.297 → **0.293** | 0.904 → **0.910** | 0.0047 → 0.0055 |
| Bear / panda | 0.175 → 0.179 | 0.779 → 0.777 | 0.764 → 0.766 | 0.235 → 0.236 | 0.899 → 0.900 | **0.0319** → 0.0323 |
| Cake / chocolate | 0.175 → 0.171 | 0.872 → 0.881 | 0.895 → 0.897 | 0.177 → 0.171 | 0.958 → 0.958 | 0.0284 → 0.0288 |
| **Mean** | 0.160 → 0.159 | 0.710 → 0.712 | 0.860 → 0.860 | 0.233 → 0.232 | 0.925 → 0.927 | 0.0176 → 0.0178 |
| Cache better | 3 (1) | 4 (2) | 5 (0) | 4 (1) | 7 (3) | 3 (0) |
| Cache-off better | 6 (0) | 5 (0) | 4 (1) | 5 (0) | 1 (1) | 6 (2) |

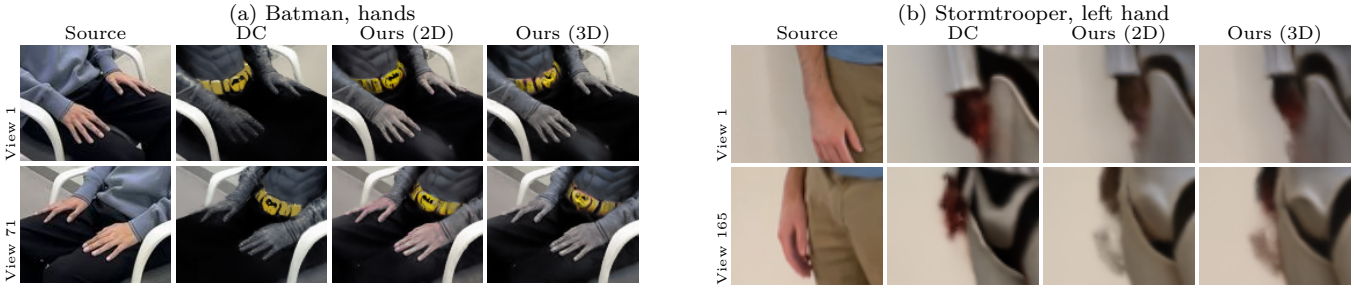



**Figure 6: Hand regions in two views of one run of each configuration.** (a) Batman: without the voxel cache (2D) the hands are gloved in view 1 and bare in view 71; with the cache (3D) they are gloved in both, as in DreamCatalyst. (b) Stormtrooper: in view 165 a faint remnant of the source hand remains beside the glove with and without the cache.

with and without the cache (Fig. 6b). An "erase-aware" anchor that releases preservation where the silhouette shrinks remains the natural extension.

**Evaluation extension:** cross-scene human preference study, to quantify when quantitative and qualitative evaluation disagree.

**Refinement stage (IV):** not evaluated, matching DreamCatalyst's own protocol. Contributions operate entirely in stage (III).

# 6 Technical Context

DreamCatalyst's DDS uses time-weighted $w_{\text{DDS}}(\hat{t}) = \delta + \gamma \hat{t}^{1/e}$ ($\delta$=0.2, $\gamma$=0.8). The target branch uses IP2P's 3-way CFG:

$$\boldsymbol{\epsilon}_{\text{tgt}} = \boldsymbol{\epsilon}_{\varnothing} + s_{\text{txt}}(\boldsymbol{\epsilon}_{\text{txt}} - \boldsymbol{\epsilon}_{\text{img}}) + s_{\text{img}}(\boldsymbol{\epsilon}_{\text{img}} - \boldsymbol{\epsilon}_{\varnothing}).$$

Our contributions operate between CFG and the DDS gradient, but mask construction and mask application occur at different points. Before any amplification, $M_{\text{tgt}}$ and $M_{\text{attn}}$ are read from the target pass, $M_{\text{src}}$ from the cached source passes, the edit strength $s$ from the CFG-combined predictions, and the edit mask $M$ (plus the voxel-cache addition $M_{\text{final}}$, from the current NeRF depth and rays) is built from them. STG and TAG then modify the target prediction. Finally, source-blended localization applies $M_{\text{final}}$ to the guided delta $\tilde{\boldsymbol{\epsilon}}_{\text{tgt}} - \boldsymbol{\epsilon}_{\text{src}}$, the preservation term is weighted by $1 - M_{\text{src}}$, and the latent-mean anchor is added. Each component can be switched off; with all of them off the pipeline is exactly DreamCatalyst.

# 7 Author Profile

I am an aspiring AI researcher who recently completed my B.Sc. in Control and Automation Engineering at UFSC (Brazil), graduating in 4.5 years, one semester ahead of the standard 5-year program (diploma pending). My interests span **generative modeling and its optimization (diffusion guidance, score distillation, 3D scene representations)**, **multimodal learning**, and **medical AI**, and I am strongly motivated to contribute to impactful research. Selected graduate-level coursework: Machine Learning (10/10), Computer Vision (10/10), Generative AI (9.5/10), Artificial Intelligence (9.5/10), and Data-Driven Algorithms (9/10). I am an AI intern at Upstart13 (US-based, remote) and previously interned at RT Medical Systems (radiotherapy/software company), where I took a medical-image segmentation pipeline from data extraction to deployment (nnU-Net to OHIF, write-up).

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